

Worksheet 1: Quantum Logic and the Pauli Algebra

Introduction to Quantum Computation

The gate toolkit. Three gates are enough for all of Part A. Qubits are numbered from 0.

- $x(a)$ — flips qubit a unconditionally. (NOT)
- $cx(a,b)$ — flips qubit b if and only if qubit a is 1. (controlled-NOT; leaves $b \mapsto a \oplus b$)
- $ccx(a,b,c)$ — flips qubit c if and only if *both* a and b are 1. (Toffoli; leaves $c \mapsto c \oplus ab$)

Write your answers as a list of gates in order, e.g. $x(0)$, $ccx(0,1,2)$, $x(2)$, and sketch the circuit if it helps. Inputs are always encoded in the lowest-numbered qubits, and any extra qubit starts in $|0\rangle$.

Matrices. $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$,
 $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$.

$M^\dagger$ denotes the conjugate transpose. The commutator is $[A, B] = AB - BA$.

Part A — Reversible logic with quantum gates

1. The Toffoli gate, and why AND needs three qubits.

- Write out the full truth table of $ccx(0,1,2)$: all eight input triples (a,b,c) and the corresponding outputs (a',b',c') .
- Every quantum gate is reversible: no two inputs may be sent to the same output. Confirm from your table that the Toffoli is reversible, and show that applying it twice returns every input unchanged.
- The classical AND gate takes two bits to one. Explain why no reversible gate can do this. *This is the reason for the convention used in Q2 and Q4:* the inputs stay where they are, and the answer is written onto a fresh output qubit that starts in $|0\rangle$. Read off from your table what the Toffoli leaves on qubit 2 when c starts at 0, and hence which classical gate $ccx(0,1,2)$ performs. (Q3 is the exception, and asks you why.)

2. Two gates to design. For each, give the gate sequence acting on three qubits, with the inputs a, b in qubits 0, 1 and the output read from qubit 2.

- The **NOR** gate: output 1 only when *both* inputs are 0.
- The **IMPLIES** gate, $a \Rightarrow b$: output 0 only when $a = 1$ and $b = 0$, and output 1 in every other case.

In each case state the Boolean identity you used (for instance, $\text{NOR}(a,b) = \bar{a} \wedge \bar{b}$), then verify your circuit against all four inputs.

3. XNOR, with one qubit fewer. The **XNOR** gate outputs 1 when its two inputs are equal.

- Give a circuit on *two* qubits only, with the output read from qubit 1.
- Explain why this gate needs no third qubit, when the two gates of Q2 did. What is it about the map $(a,b) \mapsto (a, a \oplus b)$ that the map $(a,b) \mapsto a \wedge b$ lacks?

4. A majority vote. The three-input **MAJORITY** gate outputs 1 when at least two of its three inputs are 1. This is the gate at the heart of classical error correction: given three noisy copies of a bit, it recovers the value most of them agree on.

- (a) Show that the circuit $\text{ccx}(0,1,3), \text{ccx}(1,2,3), \text{ccx}(0,2,3)$, with the output read from qubit 3, computes MAJORITY. *Hint: count how many of the three Toffolis fire when exactly k of the inputs are 1, for $k = 0, 1, 2, 3$. Remember that flipping a qubit twice does nothing.*
- (b) Why does the argument depend only on the *parity* of the number of gates that fire?

Part B — Hardware: connectivity and its cost

5. Fan-out on a real device. A real processor cannot apply cx between an arbitrary pair of qubits: only certain pairs are physically connected. To reach any other pair, the compiler inserts **swap** gates to move the information into place, and **each swap costs three cx gates**.

The five-qubit device *Belem* has the connected pairs

$$(0, 1), \quad (1, 2), \quad (1, 3), \quad (3, 4).$$

We want to run a **fan-out**: one control qubit and three targets, i.e. the three gates $\text{cx}(c, t_1), \text{cx}(c, t_2), \text{cx}(c, t_3)$. (Preceded by a Hadamard on the control, this is the circuit that creates a four-qubit GHZ state.)

- (a) Draw the connectivity as a graph — five dots, an edge for each connected pair — and write down how many neighbours each qubit has.
- (b) Find an assignment of the control and the three targets to physical qubits for which **no swap is needed at all**. How many cx gates does the circuit then cost?
- (c) Now suppose the control is placed on physical qubit 3 instead. Show that at least one **swap** is unavoidable, and give the total cx count for the best arrangement you can find. Compare with (b).
- (d) Two short ones to think about. Could this device run a fan-out onto *four* targets without a **swap**? And what is the best possible on a device whose five qubits are arranged in a straight line, $(0, 1), (1, 2), (2, 3), (3, 4)$?

This is why a processor is never described by its qubit count alone.

Part C — Bases, Paulis and conjugation

6. A basis anywhere on the Bloch sphere. On the Basics sheet, $|0\rangle$ and $|1\rangle$ were only special because we named them so. The most general single-qubit state uses two angles,

$$|0'\rangle = \cos(\theta/2) |0\rangle + e^{i\phi} \sin(\theta/2) |1\rangle,$$

where θ and ϕ are the polar and azimuthal angles on the Bloch sphere. (The lecture writes these states with bars, $|\bar{0}\rangle$; primes are used here purely because they are easier to read alongside $+$ and $-$.) The phase $e^{i\phi}$ is the new ingredient, and it has no classical counterpart at all.

- (a) Find a normalised state $|1'\rangle$ orthogonal to $|0'\rangle$, and verify $\langle 1'|0'\rangle = 0$ explicitly. *Mind the complex conjugate in the bra — with $\phi \neq 0$ it now matters.*
- (b) Write out both states for $\theta = \pi/2, \phi = \pi/2$. Identify which familiar operator has these two states as its eigenvectors, and give the corresponding eigenvalues.

7. Mutually unbiased bases. Two bases are *mutually unbiased* if a state drawn from one gives every outcome with equal probability when measured in the other.

- (a) Construct states $|+\rangle$ and $|-\rangle$ from $|0'\rangle$ and $|1'\rangle$, in the same way $|\pm\rangle$ is built from $|0\rangle$ and $|1\rangle$.
- (b) Show that $|\langle 0'|+\rangle|^2 = |\langle 1'|+\rangle|^2 = 1/2$, for *any* θ and ϕ .

- (c) Take $\theta = \pi/2$, $\phi = \pi/2$ again and write $|\pm'\rangle$ out explicitly. Which basis have you arrived at? State in one sentence what this says about measuring an eigenstate of Y in the computational basis.

8. The Pauli algebra.

- (a) Show that each of X , Y , Z is both *hermitian* ($P^\dagger = P$) and *unitary* ($P^\dagger P = I$). Deduce $P^2 = I$ without further calculation. Most matrices are neither; being both is what lets the Paulis serve as observables and as gates.
- (b) Show that $\text{tr}(P) = 0$ for each Pauli. Combined with $P^2 = I$, what does this tell you about the eigenvalues of every Pauli, without computing any?
- (c) Show that $\text{tr}(P_i P_j) = 2\delta_{ij}$ for $P_i, P_j \in \{X, Y, Z\}$.
- (d) Compute XY and YX , and hence show $[X, Y] = 2iZ$. State the other two relations obtained by cycling $X \rightarrow Y \rightarrow Z \rightarrow X$.

9. Decomposing a matrix into Paulis. Because of Q8(c), the set $\{I, X, Y, Z\}$ is an orthogonal basis for 2×2 matrices: any such M can be written $M = a_0 I + a_1 X + a_2 Y + a_3 Z$.

- (a) Show that the coefficients are given by $a_k = \frac{1}{2}\text{tr}(P_k M)$, where $P_0 = I$. *Hint: multiply the expansion by P_k and take the trace.*
- (b) Find the four coefficients for $M = \begin{pmatrix} 3 & 1-2i \\ 1+2i & -1 \end{pmatrix}$. *You can read them straight off by comparing entries — it is less work than the trace formula.*
- (c) M is hermitian. What does that imply about the four coefficients, and is it borne out by your answer?

10. Beyond the Hadamard. Notice that $H = (X + Z)/\sqrt{2}$: a combination of Paulis whose coefficients $(1, 0, 1)/\sqrt{2}$ form a *unit vector*. This suggests a whole family. For any unit vector $\hat{n} = (n_x, n_y, n_z)$ define $\hat{n} \cdot \vec{\sigma} = n_x X + n_y Y + n_z Z$.

- (a) Show that $(\hat{n} \cdot \vec{\sigma})^2 = I$ for any unit $\hat{n}$. *Hint: expand the square and use Q8 — the cross terms pair up.* Hence give its eigenvalues, and say where H sits in this family.
- (b) The phase gate $S = \text{diag}(1, i)$ does *not* square to the identity. Compute S^2 and identify it, then find the smallest positive integer n with $S^n = I$.
- (c) Compute $SXS^\dagger$, $SY S^\dagger$ and $SZ S^\dagger$, and identify each as a Pauli up to sign. A gate that maps every Pauli to a Pauli this way is called a *Clifford* gate. Using (b), describe in one sentence what S does to the Bloch sphere.

Part A built classical logic out of quantum gates and gained nothing for it — a NOR gate is a NOR gate however it is made. Part C is where the difference lives: bases that carry no information about each other, operators that fail to commute, phases with no classical counterpart. Everything from here on is built on the second half, not the first.